Bias recipe
Mathematics and implementation

Version 1 — review copy

Wenu-imaging

6 November 2026

1 Purpose and authority

The recipe estimates a master bias *signal* by correcting the electronic scan level separately in every input row, trimming the active rectangle, rejecting entire frames by their corrected active-area means, and stacking the retained frames with the existing Eclipse pixel median. QC describes the inputs, corrections and master; it does not silently modify the result.

This guide explains the implementation reviewed at cleanup-source revision `b8ca41fce6df5f34cd8668c574d873a8168797b2`. It follows the owner-approved bias/QC decisions and the final-MEF and dual-format designs listed in Section 14. Historical design documents contain superseded stages: this guide uses current source for actual arithmetic and the later approved decision for detector-mask applicability.

All signal levels, standard deviations and read-noise diagnostics are in ADU. Pixel and frame counts are dimensionless counts. The recipe performs no gain conversion, flat normalization, interpolation or dark-current estimate. It does not provide a master variance or a covariance model.

Three distinctions to retain

1. The detector BPM selects physical active pixels. It is not applied to electronic prescan or overscan samples.
2. Frame rejection selects complete exposures. The subsequent median stack still evaluates every output pixel, including pixels excluded by that BPM.
3. The historical stack median and modern QC median have different even-sample conventions. Preserving the stack is part of compatibility.

Mathematical description versus implementation

Equations below state the estimators. Exact reproduction also requires the documented order-statistic convention, evaluation order and pixel/float conversions. Checked statistics retain the existing sum-of-squares implementation; the guide does not substitute a different summation algorithm.

2 Identity, coordinates and admitted inputs

For one logical detector let $f = 0, \ldots, F-1$ identify supplied raw frames in their original order, and let $D_f(x, y)$ be a loaded value in ADU. FITS scaling belongs to the existing loader; values entering this service are the resulting `pixelvalue` samples.

Logical detector ID, source HDU, zero-based configuration index and output HDU are independent bindings. In the retained OmegaCAM run CCD65 used configuration index 0 and output HDU 1; CCD69 used index 16 and output HDU 17. The actual source HDU is checked in each input, not inferred from its ordinal.

The trim rectangle is inclusive and one-based:

$$\mathcal{R} = [x_0, x_1] \times [y_0, y_1], \qquad W_A = x_1 - x_0 + 1, \quad H_A = y_1 - y_0 + 1.$$

An active pixel (i, j) refers to raw coordinate

$$(x, y) = (x_0 + i - 1,\ y_0 + j - 1). \tag{1}$$

The code converts coordinates at the array-access boundary. This pixel identity does not validate the legacy celestial WCS or supply the deferred detector-layout fallback.

A binary BPM has an explicit HDU, polarity, coordinate domain, readout and binning. The loader normalizes the declared good value to Eclipse polarity $M = 1$ for good and $M = 0$ for excluded. Active-sized masks are embedded using Equation (1); raw-sized masks are cropped consistently for active measurements. Explicit omission is recorded; no mask is guessed from a filename.

Validation before calculation

The service checks operation/version, detector/request bindings, common image dimensions, scan/trim bounds, rejection counts, finite nonnegative frame parameter κ, positive diagnostic parameter k, and minimum input count. The configured minimum is at least three. The entire loaded raw plane must contain finite values, including masked pixels. Malformed masks or contradictory calibration declarations fail.

General statistics helpers can report nonfinite exclusions; this capability does not relax the finite-raw-input rule. No raw saturation threshold or automatic exposure discovery is added by this bias operation.

Code boundary

`mef_layout_load`, `mef_layout_load_bpm` and binding checks handle file/geometry identity. `mef_cal_bias_measurements` validates and calculates one detector using copies, leaving caller raw planes and geometry unchanged.

3 Electronic row correction

For a raw row y, concatenate the enabled prescan and overscan column bands into a vector s_y of length n_s. No detector BPM is applied. If both bands overlap, their common samples enter twice; the configuration must describe the intended electronic samples. Disabled bands contribute nothing.

Sort the samples as $s_{y,(1)} \le \cdots \le s_{y,(n_s)}$. With integer low and high rank-rejection counts ℓ, h, the row correction is

$$c_f(y) = \mathrm{pix}\left\{ \frac{1}{n_s - \ell - h} \sum_{r=\ell+1}^{n_s-h} s_{y,(r)} \right\}, \qquad n_s - \ell - h > 0. \tag{2}$$

The sum uses double precision; the result is cast to the pixel type. These are rank cuts, not iterative sigma clipping. Every column is corrected; then the active rectangle is extracted:

$$A_f(i,j) = \mathrm{pix}\{D_f(x_0 + i - 1, y_0 + j - 1) - c_f(y_0 + j - 1)\}. \tag{3}$$

The retained OmegaCAM configuration enables overscan only. CCD65 columns 2100–2142 contain 43 samples; rejecting ten at each end leaves 23. Its active rectangle is $[51, 2094] \times [5, 4095]$, or 2044×4091 pixels.

The correction profile covers every raw row, including rows outside the trim. Its legacy level is the pixel-median (Section 5); its reported RMS is a sample standard deviation about its own mean:

$$s_c = \sqrt{\frac{\sum_y (c_f(y) - \bar{c}_f)^2}{H_{\mathrm{raw}} - 1}}.$$

The implementation subtracts the median before the centered sum-of-squares evaluation. This shift preserves the exact estimator. Legacy summaries pass through float fields. Median selection reorders the working profile; scatter is invariant to reordering. The bias record stores summaries, not an ordered profile array. Separate raw scan QC uses enabled columns over trimmed rows.

`mef_overscan_correction_traced` corrects and crops; `function1d_average_reject_checked` implements Equation (2).

4 Frame admission and temporal stack

For each corrected active image, use all active pixels without a BPM, or only its good pixels otherwise. The per-frame mean is

$$a_f = \frac{1}{N_f} \sum_{(i,j)\in\mathcal{G}_f} A_f(i,j).$$

An empty valid population fails. The rejection band uses all F submitted frame means once:

$$\bar{a} = \frac{1}{F}\sum_f a_f, \qquad s_a = \sqrt{\frac{\sum_f (a_f - \bar{a})^2}{F-1}}, \tag{4}$$

$$L = \bar{a} - \kappa s_a, \qquad U = \bar{a} + \kappa s_a. \tag{5}$$

The frame-band implementation accumulates centered squared deviations in a second pass. Reject $a_f < L$ or $a_f > U$; equality is retained. The band is not recalculated after rejection. The configured minimum number of retained frames must be met. With zero scatter the band collapses to the common mean. The retained OmegaCAM setting is $\kappa = 2$, minimum three frames.

Historical configuration entries `FRAMETHR.LO`/`FRAMETHR.HI` do not add absolute cuts in this kernel. The computed band is operational. For retained set $\mathcal{K}$, $K = |\mathcal{K}|$, the master is

$$B(i,j) = \mathrm{med}_{\mathrm{Eclipse}}\{A_f(i,j) : f \in \mathcal{K}\}. \tag{6}$$

There is no per-pixel clipped mean or per-pixel rejection count. The BPM selects frame-mean and QC populations; it does not replace output pixels. The temporal median is evaluated at every active coordinate. `mef_bias_stack_masked` selects frames and consumes the private cube; `cube_avg_median` implements the stack. QC retains original input IDs.

For synthetic frame means $0, 1, 2, 10$ with illustrative $\kappa = 1$, $\bar{a} = 3.25$, $s_a = \sqrt{62.75/3} \simeq 4.573474$. The band is approximately $[-1.323474, 7.823474]$; the first three survive. This example uses $\kappa = 1$ to expose the rule, not the retained setting 2.

5 Median conventions and precision

For sorted $z_{(1)} \le \cdots \le z_{(N)}$, the modern checked QC median is

$$m(z) = \begin{cases} z_{((N+1)/2)}, & N \text{ odd}, \\ \frac{1}{2} z_{(N/2)} + \frac{1}{2} z_{(N/2+1)}, & N \text{ even}. \end{cases}$$

The Eclipse pixel median used for the temporal stack returns the *lower* middle value for even $K \ge 4$: $z_{(K/2)}$. Its two-value helper averages the pair, but the bias stack requires at least three frames. Odd-sample routines return the central order statistic. The general selection convention follows the Wirth implementation documented in the source [1].

Ten retained frames use the fifth sorted sample at each pixel; nine use the fifth. Changing this convention changes the accepted recipe. The historical `double_median` helper instead uses the upper middle for even samples; it is not the modern checked QC median.

Legacy master statistics

With a BPM, the kernel overrides the historical master median and standard deviation with checked *masked* statistics, then converts results to float. They are not inherently unmasked. Without a BPM, the historical median uses the pixel-median convention; modern QC averages the middle pair. The explicitly named `master_unmasked` family always omits the BPM. Even for equal definitions and populations, modern double results and legacy float fields can differ through rounding.

The default `pixelvalue` is float; `DOUBLEPIX` selects a different build lane. Scan sums use double precision but are converted to pixel type before subtraction. Default FITS pixels are IEEE single precision. Existing legacy cards retain six-decimal formatting; new scalar cards use 17 significant digits with a FITS-compatible uppercase exponent. A short printed table cannot establish exact numerical equality.

Worked order-statistic examples

For $0, 2, 4, 6$, modern median and mean are 3; the temporal lower median is 2. For scan samples $100, 101, 102, 103, 200$ and one rank rejected at each end, the correction is 102. Raw samples 110 and 112 become 8 and 10.

6 Checked sample statistics and MAD

A statistic defines its rectangle, processing stage, mask and nonfinite policy. Within that rectangle, BPM-excluded pixels are counted first; among remaining pixels, nonfinite samples are excluded. Thus region count equals BPM-excluded plus nonfinite-excluded plus valid count. For valid values $z_1, \ldots, z_N$,

$$\bar{z} = \frac{1}{N}\sum_i z_i, \qquad s_z = \sqrt{\frac{\sum_i (z_i - \bar{z})^2}{N-1}}, \tag{7}$$

$$d_z = m(\{|z_i - m(z)|\}), \qquad \mathrm{sMAD}(z) = 1.4826\, d_z. \tag{8}$$

Both medians in d_z use the averaged-middle convention. The raw MAD is scaled exactly once. The constant approximates $1/\Phi^{-1}(3/4)$, making scaled MAD consistent for normal-distribution standard deviation at large sample size [2, 3]. It is not a finite-sample unbiasedness correction, nor does its use establish Gaussian detector noise.

Unlike the two-pass frame-band scatter, per-pixel sample helpers evaluate

$$v = \frac{\sum_i z_i^2 - (\sum_i z_i)^2/N}{N-1}, \qquad s_z = \sqrt{\max(0, v)}.$$

They accumulate in double precision. Cancellation can affect nearly constant, large-offset data; this guide records the current arithmetic rather than replacing it with Welford accumulation. Legacy scan scatter uses the same centered sum-of-squares family after its median shift.

Situation	Representation
$N = 0$	Level and scatter unavailable; not measured zero.
$N = 1$	Mean/median measured; requested MAD is zero; sample stdev unavailable.
MAD omitted	Unavailable by explicit choice; not zero.
MAD equals zero	Valid measured result; may make a derived threshold degenerate.

The historical `image_median_stat` computes a mean absolute deviation; it must not be substituted for this median absolute deviation. `image_getstats_masked_checked` selects the population; `double_median_mad_checked` calculates checked median and MAD.

For $0, 2, 4, 6$, $s = \sqrt{20/3} \simeq 2.581989$, MAD $= 2$ and scaled MAD $= 2.9652$.

7 Which pixels belong to each diagnostic?

Family	Samples and stage	Mask
`raw_active`	Raw input, active trim rectangle	Declared detector BPM, or explicit omission
`corrected_active`	Row-corrected, trimmed input	Cropped active BPM, or omission
Prescan / overscan	Raw enabled scan band, trimmed rows	Detector BPM not applicable
Scan correction	Row corrections, full raw height	Detector BPM not applicable
`master`	Final stacked active image	Active BPM, or omission
`master_unmasked`	Same stacked active image	Explicitly no detector BPM
`raw_difference`	Original raw-pair difference, active rectangle	Intersection of both good masks

No clipping is silently added to these populations. Electronic rank rejection belongs to row correction, and whole-frame rejection belongs to stacking. Their QC context identifies these distinct stages.

Per-input scan and active measurements are retained for all submitted frames, including rejected ones. Consequently a range over all ten inputs need not summarize only the nine used for a particular detector.

Master spatial scatter describes differences across detector pixels, including structure and residual offsets. It is not the uncertainty of each master pixel. The difference between masked and unmasked scatter can be scientifically useful; neither is inherently a substitute for the other.

8 Read noise from the explicit raw pair

Two distinct, explicitly selected original raw frames a, b supply

$$\Delta(x,y) = D_a(x,y) - D_b(x,y), \qquad \mathrm{RON}_s = s_\Delta/\sqrt{2}, \quad \mathrm{RON}_{\mathrm{MAD}} = 1.4826\, d_\Delta/\sqrt{2}.$$

The population is the active rectangle with the intersection of good masks. This is measured *before* row correction or master subtraction. Selecting a raw pair is independent of later stack admission: either selected frame can be rejected from the master. `image_pair_stats_checked` forms the pixel-type difference and invokes checked sample statistics.

For independent equal-variance noise, differencing doubles the variance. More generally,

$$\mathrm{Var}(D_a - D_b) = \mathrm{Var}(D_a) + \mathrm{Var}(D_b) - 2\,\mathrm{Cov}(D_a, D_b).$$

The factor $\sqrt{2}$ therefore has an interpretation conditional on independence and equal variance. Stable spatial bias structure cancels; temporal row drift or other changing structure can contribute. Centering the difference for a standard deviation is not equivalent to electronic row correction. The generic sampled-window `image_rect_readout_noise` routine is a different estimator and is not used for this bias diagnostic.

Quantization and repeated robust values

For raw differences $-2, 0, 2, 4$, the mean is 1 and

$$s_\Delta = \sqrt{20/3}, \quad \mathrm{RON}_s = \sqrt{10/3} \simeq 1.825742, \quad d_\Delta = 2, \quad \mathrm{RON}_{\mathrm{MAD}} \simeq 2.096713.$$

Repeated values near 2.0967 ADU across detectors can result from the same integer-valued MAD of 2. They do not demonstrate identical underlying noise.

Master uncertainty is not supplied

This recipe supplies no master variance or covariance plane. In particular, RON^2/K is not its implemented uncertainty model: the temporal stack is a median, and row correction introduces shared estimates. For pixels p, q corrected using the same estimated row offset c,

$$\mathrm{Cov}(D_p - c, D_q - c) = \mathrm{Cov}(D_p, D_q) - \mathrm{Cov}(D_p, c) - \mathrm{Cov}(c, D_q) + \mathrm{Var}(c).$$

This identity explains a possible correlation; the current product does not estimate its terms or assume their independence.

9 High-offset pixels

With requested MAD, the selected master population gives median m_B and scaled MAD $s_{B,\mathrm{MAD}}$. For configured positive k,

$$T = m_B + k\, s_{B,\mathrm{MAD}}, \qquad H(i,j) = \mathbf{1}\{B(i,j) > T\}, \qquad N_H = \sum_{i,j} H(i,j).$$

The threshold population uses the declared BPM. The subsequent search visits *all* output pixels, including BPM-excluded ones, and uses strict $>$. The direct flag loop has no additional finite-value filter. Finite raw-input validation does not prove that every hypothetical floating-point overflow in later arithmetic would remain finite.

Flag polarity is 1 = flagged, unlike the normalized good-pixel BPM polarity. Coordinates are one-based in the active output. This diagnostic does not create or apply a new detector BPM. A high bias offset is not a dark-current measurement, and $k = 5$ does not by itself define a calibrated false-alarm rate.

A zero MAD is a measured but degenerate scale: the threshold becomes the median and every larger output value qualifies. With MAD omitted the derived threshold/count are unavailable. A measured zero count remains zero; an empty coordinate set is represented with its explicit availability/reason context.

For master values $0, 1, 1, 2, 20$ with no exclusions, median and MAD are both 1. At $k = 5$, $T = 1 + 5(1.4826) = 8.413$ and only 20 is flagged. If 20 is BPM-excluded, the threshold is recomputed from the remaining population, but the final search still considers 20. This distinction is deliberate.

10 FITS and ASDF: one measurement, two representations

The accepted policy retains existing FITS QC cards and writes new scalar diagnostics to both formats. Measurements and context are prepared from the same common QC record. Neither writer independently remeasures pixels. `mef_bias_qc_fits.def` is the authoritative field-to-card mapping. The following summarizes families, rather than replacing that mapping.

FITS family / field	Meaning in common QC
Legacy `FRMEDIAN`, `FRSTDEV`	Historical master statistics; retain their precision and semantics
`MAST`, `UMAS`, `DIFF`	Selected master, unmasked master, raw difference; mean, median, stdev and scaled MAD
`RON`	Raw-pair standard-deviation and MAD estimators
`HOT`	Count, threshold, multiplier and zero-scale/degeneracy context
Per-frame `RAW`, `CORR`	Raw and corrected active statistics
Per-frame `PRES`, `OVER`	Enabled electronic scan populations
Legacy scan fields	Correction-profile median and sample scatter

A statistic carries unit, status/reason, stage, region, coordinate system, mask identity/applicability, clipping and nonfinite policies, sample count and source record references as applicable. An unavailable measurement has no invented numeric card. Measured zero and unavailable are distinct.

The verified real product contained 89,445 mapped scalar/context cards: $32 \times 2795 + 5$. These are not 89,445 independent numerical diagnostics. Five detectors had one additional rejection-reason card. Arrays, embedded input records and complete provenance remain in ASDF. FITS header scalars alone do not reconstruct a full provenance tree. A future harvester must preserve units, state and definitions, not merely collect numeric keyword values.

Version domains

The retained native library version 0.3.0, ASDF file/core 1.6, application schema 2.0.0, mapping identifier `bias-scalar/1`, and Python ASDF 6 runtime refer to different version domains. They must not be compared as though they were one software version. The native interface includes the qualified memory-backed readback path for the relevant library limitation.

11 Operation, publication and explicit recovery

Normal operation keeps the raw directory untouched, launches orchestration from its own directory and writes final reduced products to the explicit product directory. In the retained run these were respectively `/Users/fselman/Pictures/OCAM/bpm`, `/Users/fselman/Pictures/OCAM/test_recipes/orchestration`, and `/Users/fselman/Pictures/OCAM/test_recipes/reduced/cal/bias`.

The standalone coordinator processes detectors serially: load and validate, measure the raw pair, correct private frame copies, compute input QC, reject frames, stack, and prepare common QC/header data. It writes private detector products and releases detector-sized arrays. A single native writer assembles the final MEF; the native QC writer publishes ASDF. A successful recipe requires the required outputs to complete. Exclusive publication prevents silent replacement; the FITS/ASDF/log set is not an atomic three-file transaction.

Expanded scalar/context headers exposed a former fixed 1 MiB aggregate readback budget. The fix budgets the exact emitted, padded header bytes with checked arithmetic while retaining the individual source-header bound. Image payload bytes are not charged to the header budget. This repairs an assembly limit; it does not change measured pixels or QC estimators.

The retained failed run was explicitly recovered from its private detector products after verification. Recovery compared all pixel payloads and legacy QC, checked mapped scalar/context cards against the original QC, then wrote a native ASDF staging file and independently validated the complete record, arrays and provenance before exclusive publication. It retained the failed log and recovery identity. It did not retroactively declare the original recipe successful or authorize unrestricted provenance harvesting.

The optional-cleanup change guards caller-side releases of absent optional objects, covers repeated/partial cleanup and creation-name replacement, and preserves allocator diagnostics for actual NULL misuse. It changes neither these equations nor the global allocation contract.

12 Implementation crosswalk

Paths below are relative to `mefpp2/`. The reviewed revision identifies actual source, while names identify the relevant calculation boundary.

Path	Responsibility
`ins/mef/common/mef_cal_bias.c`	Detector measurement service, raw RON, mask populations, hot pixels
`ins/mef/common/mef_bias_kernel.c`	Frame means, one-pass admission policy, median stack, legacy statistics
`ins/mef/common/mefp_lib.c`	Row correction, legacy scan summaries, trim and optional cleanup
`src/math/function_1d.c`	Checked rank-rejected electronic mean
`src/math/median.c`	Pixel median and order-statistic selection
`src/math/dstats.c`	Checked double median/MAD and sample scatter
`src/iproc/cube2image.c`	Temporal per-pixel median
`src/iproc/image_stats.c`	Checked masked/pair populations and statistics
`ins/mef/common/mef_bias_product_qc.c`	Shared measurement/context record construction
`ins/mef/common/mef_bias_qc_fits.c` and `mef_bias_qc_fits.def`	Scalar/context FITS serialization and authoritative mapping
`ins/mef/common/mef_bias_files.c`	File adapter and detector-product preparation
`ins/mef/common/mef_operation_template.c`	Operation lifetime and cleanup
`ins/mef/common/mef_mef_assemble.c`	Native MEF assembly and header budget
`ins/mef/common/mef_product_qc_asdf.c`	Native ASDF writing, readback and explicit recovery
`ins/mef/recipes/mefpp_bias_recipe.c`	Standalone serial coordinator

13 Qualification and limits of the evidence

The retained Mac exercise preserved ten original OmegaCAM bias frames and produced 32 detector extensions. Its comparison established:

- 864 legacy QC keyword groups matched retained detector references exactly;
- all 32 recovered detector pixel payloads matched the earlier mosaic byte for byte; recovered FITS size was 1,077,825,600 bytes;
- all 89,445 mapped scalar/context cards agreed with the original QC;
- independent ASDF checks validated the complete recovered record, including schema, values, array content/float bits, provenance and FITS mapping;
- focused Mac checks passed for aggregate header budgeting, individual header bounds, optional cleanup and four in-memory QC profiles.

The existing recipe was relinked with accurate revision stamp `c1751306f817d37b9c4791d65f8925f57ef1151b`. Cleanup-source revision `b8ca41fce6df5f34cd8668c574d873a8168797b2` was separately component-qualified; that result does not imply the production recipe was subsequently relinked.

Evidence receipts are retained under the selected Mac qualification directory: `scalar-qc-qualification-c17/qualification.json`, `header-budget-checks/mef-assembly-results.json`, `recipe-relink.json` and `optional-cleanup-results.json`. Recovery receipt: orchestration file `bias-scalar-qc-asdf-recovery.json`. See repository status notes `bias-scalar-qc-2026-11-06.md` and `native-optional-cleanup-2026-11-06.md` for the staged history.

These establish the reported comparisons and focused Mac behavior. Historical Mac/Chapman core comparisons do not qualify every new native-ASDF path on all platforms. This is not a claim of detector health acceptance, ESO Phase-3 compliance, general out-of-memory coverage or reentrancy.

Two nonstandard lowercase-exponent telemetry cards were byte-identical in raw and product primary headers. Structural/QC agreement does not resolve their strict FITS conformance. This inherited header issue is separate from the bias arithmetic and the corrected aggregate assembly limit.

This guide changes documentation only. It requires no raw replay, compiler selection, build-directory creation or CMake reconfiguration. Future code edits must reuse the selected build/toolchain unless a configuration change is intentional and explained.

14 Approved design basis

Repository paths identify the human-reviewed design lineage. Later decisions supersede earlier scope or serialization arrangements where stated.

1. `docs/architecture/proposed/m1/bias-service.md`: numerical compatibility, operation boundary and ownership.
2. `docs/developer/bias-qc-eclipse-reuse-review-2026-11-02.md`: reused statistics, explicit raw-pair RON and high-offset diagnostics.
3. `docs/developer/bias-bpm-asdf-handoff-2026-11-02.md`: detector BPM applies to active populations, not electronic scan samples.
4. `docs/developer/design/bias-mef-native-asdf-v1.md`: final native MEF/ASDF product and raw/orchestration/reduced-directory operation.
5. `docs/developer/design/bias-qc-fits-asdf-mapping-v1.md`: retain legacy cards and serialize new scalar diagnostics to both formats.

This is the post-implementation mathematical account for owner review, not an amendment to the accepted numerical policy. Source discrepancies discovered in future review should be resolved explicitly rather than hidden by changing an equation or a test expectation.

References

[1] N. Wirth, *Algorithms + Data Structures = Programs*, Prentice-Hall, 1976. Algorithm attribution as documented by the retained Eclipse median source; exact even-sample conventions are stated in this guide.

[2] R Core Team, *Median Absolute Deviation*, R stats reference manual, accessed 6 November 2026. https://www.stat.math.ethz.ch/R-manual/R-patched/RHOME/library/stats/html/mad.html. Normal-consistency scaling and median conventions.

[3] NIST, *Median Absolute Deviation*, Dataplot reference manual, accessed 6 November 2026. https://www.itl.nist.gov/div898/software/dataplot/refman2/auxillar/mad.htm. Definition of MAD and normal-scale interpretation.